This is going to be a broad overview of the ideas and techniques used to prove the following much celebrated theorem.

Theorem (Gromov's Theorem on Groups of Polynomial Growth)**:** If a finitely generated group Γ has polynomial growth, then Γ contains a nilpotent subgroup of finite index.

All of this work is discussed in Gromov's paper and the paper by van den Dries and Wilkie, and I am only writing this up for personal use (and to present to the University of Virginia Graduate Students Seminar).

Necessary Algebraic Results

Definition: Suppose Γ has polynomial growth. We will let growth(Γ) be the smallest natural number d such that there exists C such that $|B(r)| \leq Cr^d$ for each $r \geq 1$. Here, $B(r)$ denotes the closed unit ball of radius r centered at the identity.

Suppose $\Gamma' \leq \Gamma$ is a subgroup of a finitely generated group acting via translation. Let $X = \Gamma/\Gamma'$ be the quotient space, and let $f : \Gamma \to X$ the natural projection. Let $\text{dist}(x,y) = \inf_{\alpha,\beta} d(\alpha,\beta)$ for any $\alpha \in f^{-1}(x)$ and $\beta \in f^{-1}(y)$. Since Γ' acts via isometries on Γ, it follows that $\text{dist}(x,y)$ is a metric on X.

Observe that for any two points $x, y \in X$ with $\text{dist}(x,y) = p$ for some integer p, there exist points $x = x_0, x_1, \ldots, x_p = y$ such that $\text{dist}(x_i, x_{i-1}) = 1$. Importantly, this means that if X is infinite, then $|B_X(x,r)| \geq r+1$ for all $x \in X$ and all $r \geq 0$.

Lemma (Splitting Lemma)**:** If $\Gamma' \subseteq \Gamma$ is a *finitely generated* subgroup of *infinite* index. Then, growth(Γ') $\leq$ growth(Γ) $- 1$.

Proof. Since Γ/Γ' is connected, it follows that $B(r)$ for each $r \in \Gamma$ contains at least $r+1$ elements $\alpha_0, \ldots, \alpha_r \in \Gamma$ such that $f(\alpha_i) \neq f\left(\alpha_j\right)$ whenever $i \neq j$, where $f : \Gamma \to \Gamma/\Gamma'$ is the quotient map. The intersections $B' = B(r) \cap \Gamma'$ and the translates $B'\alpha_i$ are disjoint, and they are contained in $B(2r)$, so that $|B(2r)| \geq (r+1)|B'|$. □

The following theorems are especially important in the algebraic proof. We will not prove them here.

Theorem (Jordan's Theorem)**:** For any Lie group L with finitely many components, there is a number q such that every finite subgroup in L contains an abelian subgroup of index at most q.

Theorem (Tits Alternative)**:** Let L be a Lie group with finitely many components, and let $G \subseteq L$ be an arbitrary finitely generated subgroup. Then, either

- G has a free group of rank 2 (hence is of exponential growth);
- G is virtually solvable (hence G has exponential growth unless it is virtually nilpotent).

Theorem (Milnor–Wolf Theorem)**:** If Γ is a virtually solvable group, then either Γ is of exponential growth or Γ is virtually nilpotent.

Remark: Given a nilpotent group G with lower central series

$$G = \gamma_1(G) \geq \gamma_2(G) \geq \cdots \geq \gamma_c(G) \geq \gamma_{c+1}(G) = \{e_G\},$$

the Bass–Guivarch formula gives that G grows polynomially with degree given by

$$d = \sum_{i=1}^{c} i \,\text{rk}(\gamma_i(G)/\gamma_{i+1}(G)),$$

where rk denotes the rank of the abelian groups $\gamma_i(G)/\gamma_{i+1}(G)$.

Lemma (Lemma (a))**:** Let L be a Lie group with finitely many components, and suppose that for each $p = 1, 2, \ldots$, there is a homomorphism $G \to L$ such that its image is finite and has at least p elements. Then, G contains a subgroup $G' \subseteq G$ of finite index such that $[G', G'] \subseteq G'$ has infinite index.

Proof. We start by observing that if G is a finitely generated group with finite generating set S, then given any index k, G has finitely many subgroups of index k. This follows from the fact that if H is any such sub-

group, then the action of G on G/H yields a homomorphism $G \to S_k$ (and so H is determined by the given homomorphism), of which there are at most $(k!)^{|S|}$ such homomorphisms.

Therefore, there are finitely many subgroups of index at most q, and since the finite intersection of finite index subgroups is finite index, it follows that if we let

$$G' \coloneqq \bigcap \{H \leq G : [G:H] \leq q\},$$

then G' is of finite index.

We will show that $[G', G']$ is of infinite index by showing that G' admits a surjection onto abelian subgroups of arbitrary size. If $\phi\colon G \to L$ has image $\left|F_p\right| \geq p$, then $G' \subseteq \phi^{-1}\left(A_p\right)$, where $A_p \leq F_p$ is abelian and has index q. Since

$$\left[F_p : \phi(G')\right] \leq [G : G'],$$

it follows that

$$\left|\phi(G')\right| \geq \frac{p}{[G:G']}.$$

Since p is arbitrary, it follows that $[G', G'] \subseteq G'$ has infinite index. □

Lemma (Lemma (b))**:** Suppose Γ is a finitely generated group of polynomial growth. Then, $[\Gamma, \Gamma]$ is finitely generated.

Proof. It suffices to show that the kernel $\Delta \subseteq \Gamma$ of any homomorphism $g\colon \Gamma \to \mathbb{Z}$ is finitely generated.

Let $\gamma_0, \gamma_1, \ldots, \gamma_k \in \Gamma$ be a system of generators such that $g(\gamma_0) = 1 \in \mathbb{Z}$ and $\gamma_i \in \Delta$ for each $i = 1, \ldots, k$. Let

$$\Delta_m = \left\langle \left\{ \gamma_0^j \gamma_i \gamma_0^{-j} : 1 \leq i \leq k, -m \leq j \leq m \right\} \right\rangle.$$

Then,

$$\Delta = \bigcup_{m=0}^{\infty} \Delta_m.$$

If the sequence of Δ_m stabilizes, then it equals Δ and we are done. Else, there are $\alpha_m \in \Delta$ such that each α_m is of the form $\alpha_m = \gamma_0^m \gamma_i \gamma_0^{-m}$ or $\alpha_m = \gamma_0^{-m} \gamma_i \gamma_0^m$ satisfying $\alpha_m \notin \langle \alpha_0, \alpha_1, \ldots, \alpha_{m-1} \rangle$.

Were such a family to exist, our goal is to show that it implies the existence of an exponentially growing family. Let $\beta(\epsilon_0, \ldots, \epsilon_m) = \alpha_0^{\epsilon_0} \cdots \alpha_m^{\epsilon_m}$, where $\epsilon_i \in \{0, 1\}$ for each i. Since equality between any two such elements β is defined by equality of each ϵ_i, it follows that there are 2^{m+1} such β.

Notice then that $\ell(\beta) \leq \sum_{i=1}^{m} \ell(\alpha_m) \leq (m+1)(2m+1)$. Thus, $|B((m+1)(2m+1))| \leq 2^{m+1}$ as the ball of radius $(m+1)(2m+1)$ has all the β. Yet, this contradicts polynomial growth. □

The most important applications of these are to prove the following lemma.

Lemma (Lemma 1)**:** Let Γ be a finitely generated group of polynomial growth, and let L be a Lie group with finitely many connected components. Suppose that for each finitely generated infinite group $\Gamma' \subseteq \Gamma$, there is a subgroup $\Delta \subseteq \Gamma'$ of finite index such that, for all $p = 1, 2, \ldots$, there is a homomorphism $\Delta \to L$ such that its image contains at least p elements.

Then, Γ is virtually nilpotent.

Proof. By the splitting lemma, we inductively assume that all finitely generated subgroups of Γ with infinite index are virtually nilpotent.

Let $\Delta \subseteq \Gamma$ be a subgroup of finite index with the required homomorphisms into L. If these homomorphisms are all finite, Lemma (a) yields a subgroup $\Delta' \subseteq \Delta$ with finite index such that $[\Delta', \Delta'] \subseteq \Delta$ has infinite in-

dex. Induction yields the final result in this case.

If there is a homomorphism $\Delta \to L$ with infinite image, then the Tits theorem applied to this image yields a finite index subgroup $\Delta' \subseteq \Delta$ such that $[\Delta', \Delta'] \subseteq \Delta$ is infinite index as well. According to Lemma (b), the commutator $[\Delta', \Delta']$ is finitely generated, so by the induction hypothesis, it is virtually nilpotent. Therefore, Γ is virtually solvable, so by the Milnor–Wolf Theorem, it follows that Γ is virtually nilpotent. □

Notions from Nonstandard Analysis

In Gromov's original paper, he constructed a special metric space Y as a subsequential limit of the sequence of discrete metric spaces $(\Gamma, (1/n)d_S)$, where S is a generating set for S. The metric space Y has the following properties:

(i) Y is homogeneous;

(ii) Y is connected and locally connected;

(iii) Y is complete;

(iv) Y is locally compact and of finite topological dimension.

The deep result of Gleason–Montgomery–Zippin gives that $\mathrm{Isom}(Y)$ is then a connected Lie group. There is then a finite-index subgroup $\Gamma' \subseteq \Gamma$ that admits a homomorphism $\Gamma' \to \mathrm{Isom}(Y)$ such that the image of Γ' has arbitrary order, so we may use Lemma 1 to obtain the desired result.

Definition: Given a metric space Y with a distinguished point e, $\mathrm{Isom}(Y)$ is topologized by defining the basic closed neighborhoods

$$B_{k,\varepsilon} = \{\sigma \in \mathrm{Isom}(Y) : k \in \mathbb{Z}_{>0} \text{ and } \varepsilon > 0 \text{ with } d(\sigma(y), y) \leq \varepsilon \text{ for all } y \text{ with } d(y, e) \leq k\}$$

at the identity.

The way we will construct this is through the asymptotic cone. We will use the construction in [DW84] that involves methods from nonstandard analysis.

Let $\mathbb{R}$ denote the real numbers, and let $\mathcal{U}$ be an ultrafilter on $\mathbb{N}$. On the set $\mathbb{R}^{\mathbb{N}}$, we may define an equivalence relation by saying two sequences $(u_n)_n$ and $(v_n)_n$ are equivalent if

$$\{n \in \mathbb{N} : u_n = v_n\} \in \mathcal{U}.$$

We denote by $\mathbb{R}^{\mathcal{U}}$, which will be written as $\mathbb{R}^*$, the ultrapower of $\mathbb{R}$. It c an be shown that the operations that turn $\mathbb{R}$ into an ordered field, such as addition, subtraction, multiplication, division, and the ordering on $\mathbb{R}$ admit extensions to $\mathbb{R}^*$ that turn $\mathbb{R}^*$ into an ordered field. We will identify $\mathbb{R} \subseteq \mathbb{R}^*$ by taking $s \mapsto [s]$, where $[s]$ is the equivalence class of the constant sequence with value s.

In particular, we define

$$\begin{aligned}
[(u_n)_n] \pm [(u_n)_n] &= [(u_n \pm v_n)_n] \\
[(u_n)_n][(v_n)_n] &= [(u_n v_n)_n] \\
[(u_n)_n]/[(v_n)_n] &= \begin{cases} u_n/v_n & v_n \neq 0 \\ 0 & v_n = 0, \end{cases}
\end{aligned}$$

and we say that $[(u_n)_n] \leq [(v_n)_n]$ if and only if $\{n : u_n \leq v_n\} \in \mathcal{U}$.

We note that the nonstandard construction does not preserve the Archimedean property. In particular, we may define an element $[u]$ to be *finite* if there exists $s \in \mathbb{R}$ with $s > 0$ such that $[-s] \leq [u] \leq [s]$, and we say the element is *infinite* if it is not finite. We say the element is *infinitesimal* if for all $s \in \mathbb{R}$ with $s > 0$, we have $[-s] \leq [u] \leq [s]$. If $\mathbb{R}^{\mathrm{fin}}$ denotes the finite elements of $\mathbb{R}^*$ and $\mathbb{R}^0$ denotes the infinitesimal elements, then there is an isomorphism $\rho\colon \mathbb{R} \to \mathbb{R}^{\mathrm{fin}}/\mathbb{R}^0$; if $h\colon \mathbb{R} \to \mathbb{R}^{\mathrm{fin}}$ is the natural inclusion map, then the homomorphism $\mathrm{st} = \rho^{-1} \circ h$ from $\mathbb{R}^{\mathrm{fin}}$ to $\mathbb{R}^0$ is called the *standard part* map.

Definition: Let S be any set, ω an ultrafilter on $\mathbb{N}$, and let S^ω be the corresponding ultrapower. That is, $S^\omega = S^{\mathbb{N}}/\sim_\omega$, where $f, g \in S^{\mathbb{N}}$ are equal if one has $\{i \in \mathbb{N} : f(i) = g(i)\} \in \omega$. Identify $S \subseteq S^\omega$ with the equivalence classes of their respective constant functions.

Given sets $S_1, \ldots, S_n$, a subset $W \subseteq S_1^\omega \times \cdots \times S_n^\omega$ is called *internal* if it is realized by coordinates. That is, for each $k \in \mathbb{N}$, there is $W_k \subseteq S_1 \times \cdots \times S_n$ such that for all $f_1 \in S_1^{\mathbb{N}}, \ldots, f_n \in S_n^{\mathbb{N}}$, we have that

$$([f_1], \ldots, [f_n]) \in W$$

if and only if

$$(f_1(k), \ldots, f_n(k)) \in W_k \text{ for a.e. } k.$$

Similarly, a map $F : S_1^\omega \times \cdots \times S_{n-1}^\omega \to S_n^\omega$ is called internal if for all $k \in \mathbb{N}$, there exists $F_k : S_1 \times \cdots \times S_{n-1} \to S_n$ such that for all $f_1 \in S_1^{\mathbb{N}}, \ldots, f_{n-1} \in S_{n-1}^{\mathbb{N}}$, we have that

$$F([f_1], \ldots, [f_{n-1}]) = \left[(F_k(f_1(k), \ldots, f_{n-1}(k)))_k\right]$$

Notice that internal subsets are closed under intersections, unions, and set subtraction.

We will need the following lemmas related to internal subsets and functions on the nonstandard naturals.

Lemma: Suppose $g_n \in S^\omega$ for each $n \in \mathbb{N}$. Then, there is an internal function $F : \mathbb{N}^* \to S^\omega$ such that $F(n) = g_n$ for all $n \in \mathbb{N}$.

Proof. Let $g_n = [f_n]$ for some function $f \in S^{\mathbb{N}}$. For each $i \in \mathbb{N}$, let $F_i : \mathbb{N} \to S$ be defined by $F_i(n) = f_n(i)$. Then, we define $F : \mathbb{N}^* \to S^\omega$ to be the map with components $\{F_i : i \in \mathbb{N}\}$. For any $n \in \mathbb{N}$, we identify $\hat{n} \in \mathbb{N}^*$ with the constant sequence. Then, we have

$$\begin{aligned} F(\left[(\hat{n}(i))_{i\in\mathbb{N}}\right]) &= \left[(F_i(\hat{n}(i)))_{i\in\mathbb{N}}\right] \\ &= \left[(F_i(n))_{i\in\mathbb{N}}\right] \\ &= \left[(f_n(i))_{i\in\mathbb{N}}\right] \\ &= g_n. \end{aligned}$$

□

Lemma: Suppose $A \subseteq \mathbb{N}^*$ is an internal subset. Then, A has a least element.

Proof. Write $(A_n)_{n\in\mathbb{N}}$ for the components of A. Then, the equivalence class of the sequence $(\min(A_n))_{n\in\mathbb{N}}$ is the smallest element of A. □

Lemma (Overspill Lemma)**:** Suppose $X \subseteq \mathbb{N}^*$ is an internal subset, and $k \in \mathbb{N}$. If $n \in X$ for all $n \in \mathbb{N}$ with $n \geq k$, then there is an infinite $K \in \mathbb{N}^*$ with $n \in X$ for all $n \in \mathbb{N}^*$ with $k \leq n \leq K$.

Proof. Suppose it were the case that all infinite nonstandard naturals were in K. Then, any infinite $K \in \mathbb{N}^*$ is satisfactory.

Else, there are infinite nonstandard naturals not in X. Our goal is now to show that there is a smallest such infinite nonstandard natural not in X. It would then be the case that all infinite nonstandard naturals smaller than H will be in X.

We observe that the set

$$Y = \{n \in \mathbb{N}^* : k < n \in \mathbb{N}^* \setminus X\}$$

is then nonempty, and Y is internal since Y is given by

$$(\mathbb{N}^* \setminus \{1, \ldots, k\}) \cap (\mathbb{N}^* \setminus X).$$

Therefore, Y has a least element H. So, H is a nonstandard natural greater than k but not in X; since all finite $n \geq k$ are in X, it follows that H is infinite. Furthermore, $K = H - 1$ then is infinite and satisfies the requirements of the theorem. □

The Los theorem says it is possible to show that a first order sentence Φ^ω in elements and internal subsets in the ultrapowers $S_1^\omega \times \cdots \times S_n^\omega$ if and only if the underlying first order sentence Φ holds for almost every $i \in \mathbb{N}$. In fact, if Φ is an elementary statement (i.e., expressed entirely in elements), then Φ^* expresses the same property as Φ.

Constructing the Asymptotic Cone Y

Let ω be an ultrafilter on $\mathbb{N}$. Then, analogous to how we constructed $\mathbb{R}^*$, we are able to construct the ultrapower Γ^ω by taking $\Gamma^\omega/\sim_\omega$. We note that if S is a finite generating set for Γ, then there is a statement in elements and sets that says "S generates Γ," so there is a nonstandard extension. Every element $g \in \Gamma^\omega$ may then be written as a word in the generators, but the length of the word may be a nonstandard natural number (i.e., it may be infinite); since the set of all $n \in \mathbb{N}^*$ for which such a word of length n exists is an internal subset (for fixed g), it follows that there is a nonstandard length function, which we will denote ℓ_ω.

Let $R \in \mathbb{R}^*$ be an infinite nonstandard real. Define

$$\Gamma^{(R)} = \left\{ g \in \Gamma^\omega : \text{there exists } A > 0 \text{ in } \mathbb{R} \text{ such that } \frac{\ell_\omega(g)}{R} \leq A \right\}$$

as a subgroup of Γ^ω. Let

$$N = \left\{ g \in \Gamma^\omega : \frac{\ell_\omega(g)}{R} \text{ is infinitesimal} \right\}$$

be the subgroup of infinitesimal elements of $\Gamma^{(R)}$. Then, $\ell_\omega(\cdot)/R$ defines a map from $\Gamma^{(R)}$ to $\mathbb{R}^{\text{fin}}$, and $\ell_\omega(g)/R - \ell_\omega(h)/R$ is infinitesimal whenever $gN = hN$ in the coset space $\Gamma^{(R)}/N$. Therefore, the map ℓ_ω descends to a map $\overline{\ell}\colon \Gamma^{(R)}/N \to \mathbb{R}$ defined by

$$\overline{\ell}(gN) = \text{st}(\ell_\omega(g)/R).$$

Let $Y = \Gamma^{(R)}/N$. Then, Y is equipped with the metric $\overline{d}$ given by $\overline{d}(gN, hN) = \overline{\ell}\left(h^{-1}gN\right)$.

Proposition: The metric space Y is

(i) homogeneous;

(ii) connected and locally connected (by geodesics);

(iii) complete.

Proof.

(i) The group $\Gamma^{(R)}$ acts on Y from the left by isometries, with for all $a \in \Gamma^{(R)}$ having

$$\begin{aligned} \overline{d}(agN, ahN) &= \overline{\ell}\left(h^{-1}a^{-1}agN\right) \\ &= \overline{\ell}\left(h^{-1}gN\right) \\ &= \overline{d}(gN, hN). \end{aligned}$$

This action is transitive. Thus, Y is homogeneous.

(ii) Without loss of generality, assume $p = eN$ and $q = gN$. Furthermore, we may assume without loss of generality that $\overline{\ell}(gN) = 1$ (extending by concatenating paths), meaning that $\ell_\omega(g) = R$. We may extend the floor function $\lfloor \cdot \rfloor$ to $\mathbb{R}^*$. We may then write g as a representation

$$g = g_1 \cdots g_{\lfloor R \rfloor}$$

as a nonstandard word. The map $f : [0,1] \to Y$ given by $f(t) = g_1 \cdots g_{\lfloor tR \rfloor}$ is then continuous.

(iii) Suppose $(g_nN)_{n\in\mathbb{N}}$ is a Cauchy sequence in Y. Assume that this Cauchy sequence is bounded by 1. Therefore, we have that $\ell_\omega(g) \le R$.

We may extend $(g_n)_{n\in\mathbb{N}}$ to an internal sequence $(g_n)_{n\in\mathbb{N}^*}$. For each strictly positive (standard) natural number k, let $M(k) \in \mathbb{N}$ be such that $\ell_\omega\left(g_m^{-1}g_n\right) < R/k$ for all (standard) naturals $m, n \ge M(k)$. This may be extended to all nonstandard natural numbers $m, n \in \mathbb{N}^*$ bounded above by some infinite $N(k) \in \mathbb{N}^*$. This follows from the application of the above argument to the internal set

$$\left\{t \in \mathbb{N}^* : \text{ for all } m,n \in \mathbb{N}^*, \text{ one has that } M(k) < m,n < t \text{ implies that } \ell_\omega\left(g_m^{-1}g_n\right) < R/k\right\}.$$

So, there is some $\theta \in \mathbb{N}^*$ such that $\theta \ge M(k)$ and $\theta \le N(k)$ for all standard natural numbers $k > 0$. Thus, we have $\lim_n g_nN = g_\theta N$ in Y.

□

It is important to note that the space Y depends on the choice of nonstandard real R that is used to define it. However, it is not dependent on the choice of generating set, since any two choices of generating sets admits a bilipschitz equivalence (by writing generators in one set through generators in the other set).

Local Compactness and Finite-Dimensionality of Y

From now on, we will write d for the distance metric on Y and ℓ for the length function on Γ^ω.

Lemma: Let R_0 be a positive infinite nonstandard real, and suppose d is such that $|B_Y(e,R_0)| \le cR_0^d$ for some positive finite real number c and some natural $d \in \mathbb{N}$.

Then, there is a positive infinite $S \le R_0$ such that for every $i \in \mathbb{N}$ with $i \ge 4$, the following property denoted $P_i(S)$ holds: if $g_1,\dots,g_t \in B_Y(e,S/4)$ for some $t \in \mathbb{N}^*$, and $B_Y(g_1,S/i),\dots,B_Y(g_t,S/i)$ are pairwise disjoint, then $t \le i^{d+1}$.

Proof. Suppose the lemma is false. Then, for all $S \in \mathbb{R}^*$ with $\log(R_0) \le S \le R_0$, there is some $i \ge 4$ in $\mathbb{N}$ such that $P_i(S)$ fails.

The map $S \to \min\{i \in \mathbb{N} : P_i(S) \text{ fails}\}$ is an internal function whose range is in the naturals, so it is finite and hence bounded by some $K \in \mathbb{N}$.

By internal induction, we may define natural numbers $4 \le i_1,\dots,i_u \le K$ for some $u \in \mathbb{N}^*$ and elements $g(l,j) \in \Gamma^\omega$, where $1 \le l \le u$, $1 \le j \le \lfloor i_l^{d+1} \rfloor + 1$, such that

- $g(l,j) \in B_Y\left(e, \frac{R_0}{4i_1\cdots i_{l-1}}\right)$, and
- $B_Y\left(g(l,j), \frac{R_0}{i_1\cdots i_{l-1}i_l}\right) \cap B_Y\left(g(l,j'), \frac{R_0}{i_1\cdots i_{l-1}i_l}\right) = \emptyset$ for each $1 \le j < j' \le t_l$.

We note that this inductive definition of i_l and $g(l,j)$ may continue as long as the condition $P_i(R_0/(i_1\cdots i_{l-1}))$ is not satisfied. Such a family will be guaranteed to exist if we choose u such that

$$\frac{R_0}{i_1\cdots i_{u-1}} \ge \log(R_0) > \frac{R_0}{i_1\cdots i_{u-1}i_u}.$$

Let

$$T = \{(s_1,\dots,s_u) : s_l \in \mathbb{N}, 1 \le s_l \le t_l \text{ for each } l = 1,\dots,u\}.$$

For any $s = (s_1,\dots,s_u) \in T$, define

$$g_s = g(1,s_1)g(2,s_2)\cdots g(u,s_u),$$

which satisfies

$$\ell(g_s) \le \sum_{l=1}^{u} \ell(g(l,s_l))$$

$$\leq \sum_{l=1}^{u} \frac{R_0}{4i_1 \cdots i_{l-1}}$$
$$\leq \sum_{l=1}^{u} \frac{R_0}{4^l}$$
$$< R_0.$$

Therefore, we have the containment $\{g_s : s \in T\} \subseteq B(e, R_0)$. If $s \neq s'$ in T, then we claim that $g_s = g_{s'}$. If not, then there is some $v < u$ such that

$$g(v, s_v) \cdots g(u, s_u) = g(v, s'_v) \cdots g(u, s'_u)$$

where $s_v \neq s'_v$. Then, it follows that

$$g(v, s'_v)^{-1} g(v, s_v) = g(v+1, s'_{v+1}) \cdots g_{u,s'_u} g(u, s_u)^{-1} \cdots g(v+1, s_{v+1})^{-1}.$$

Yet, this means that

$$\frac{R_0}{i_1 \cdots i_v} \leq \ell\Big(g(v, s'_v)^{-1} g(v, s_v)\Big)$$
$$\leq 2 \sum_{l=v+1}^{u} \frac{R_0}{4i_1 \cdots i_{l-1}}$$
$$\leq \frac{R_0}{2i_1 \cdots i_v}\left(1 + \sum_{l=v+2}^{u} \frac{1}{4^{l-v-1}}\right)$$
$$< \frac{R_0}{i_1 \cdots i_v}.$$

Thus, $g_s \neq g_{s'}$ for any $s \neq s'$. Yet, this means that

$$|\{g_s : s \in T\}| = |T|$$
$$\geq \prod_{l=1}^{u} i_l^{d+1}$$
$$> (R_0/\log(R_0))^{d+1}$$
$$> cR_0^d,$$

where the latter inequality emerges from the fact that R_0 is infinite. Thus, we have that $|B(e, R_0)| > cR_0^d$, yielding our desired contradiction. □

Now we will briefly discuss topological dimension. Let $p \in \mathbb{R}$ be any positive real number, and let $\varepsilon > 0$. Suppose X is a topological space. We let

$$H_p^{\delta}(X) = \inf\left\{ \sum_{j=1}^{\infty} \Big(\operatorname{diam}\Big(U_j\Big)\Big)^p : X = \bigcup_{j=1}^{\infty} U_j \text{ and } \operatorname{diam}\Big(U_j\Big) < \delta \text{ for each } j \right\}.$$

Then, we let

$$H_p(X) = \sup_{\delta > 0} H_p^{\delta}(X).$$

This is the *Hausdorff content* of X. We define the *dimension* of X to be the supremum over all $p \geq 0$ such that $H_p(X) > 0$.

For the following proposition, we will let $Y^{(R)}$ denote the space $\Gamma^{(R)}/N$ from earlier.

Proposition: Suppose growth$(\Gamma) \leq d$ for some $d \in \mathbb{N}$. Then, there exists some positive infinite $R \in \mathbb{R}^*$ such that the metric space $Y^{(R)}$ is locally compact with (Hausdorff) dimension $d \leq 1$.

Proof. By the overspill principle, there is some positive infinite $R_0 \in \mathbb{R}^*$ such that $|B(e, R_0)| \leq cR_0^d$. Let $S \in \mathbb{R}^*$ be such that the property $P_i(S)$ from the previous lemma holds. Set $R = S/4$. We will show that $Y^{(R)}$ satisfies the necessary conditions.

Let $k \in \mathbb{N}$ be strictly positive. In particular, by the way that $P_i(S)$ is defined, we have that for any $g_1, \ldots, g_t \in B(e, R)$ with $t \in \mathbb{N}^*$ such that the balls $B(g_1, R/k), \ldots, B(g_t, R/K)$ are disjoint, then $t \leq (4k)^{d+1}$. That is, t is finite. Taking a maximal such t, the balls $B(g_i, 2R/k)$ cover $B(e, R)$, as else it would follow that $B(g, R/k)$ would be disjoint from all the $B(g_i, R/k)$, contradicting maximality.

The coset map $\Gamma^{(R)} \to Y^{(R)}$ gives that the closed ball of radius 1 and center eN in $Y^{(R)}$ is covered by at most $(4k)^{d+1}$ balls of radius $2/k$. In particular, since $Y^{(R)}$ is homogeneous and complete, we have that all closed balls of Y with radius 1 are compact (i.e., they're necessarily totally bounded), so that they are of Hausdorff dimension at most $d+1$. Thus, $Y^{(R)}$ is locally compact of dimension at most $d+1$. □

Therefore, by the Gleason–Montgomery–Zippin Theorem (or rather, a corollary thereof), we have that Isom(Y) is a Lie group with finitely many connected components.

The Action of Γ on Y

We know that left multiplication yields a homomorphism $l \colon \Gamma^{(R)} \to \text{Isom}(Y)$, and from the previous section, that Isom(Y) is a Lie group with finitely many components.. Our next job is to establish the existence of a finite-index subgroup Γ' such that Γ' admits homomorphisms into Isom(Y) of arbitrary degree. An application of Lemma 1 will then yield the desired result.

We will assume that $l(\Gamma) \subseteq \text{Isom}(Y)$ be finite, and let $\Gamma' = \ker(l|_\Gamma)$. In particular, Γ' is of finite index in Γ, and there is some finite symmetric generating set $S \subseteq \Gamma'$.

Lemma: If Γ has no finite index abelian subgroups, then the set of lengths $\left\{\ell\left(\gamma^{-1} s \gamma\right) : s \in S, \gamma \in \Gamma'\right\}$ is unbounded.

Proof. Otherwise, each s has only finitely many Γ'-conjugates, meaning that the $C_{\Gamma'}(s)$ is finite index in Γ'. Since the center of Γ' is the intersection of the centralizers of each of the $s \in S$, it follows that this center is finite index in Γ', hence finite index in Γ. □

Recall that the group Isom(Y) is topologized by the basic closed neighborhoods

$$B_{k,\varepsilon} = \{\sigma \in \text{Isom}(Y) : d(\sigma(y), y) \leq \varepsilon \text{ for all } y \text{ with } d(y, e) \leq k\}.$$

at the identity for any $0 < \varepsilon \in \mathbb{R}$ and $0 < k \in \mathbb{N}$.

Proposition: Suppose Γ has no abelian subgroups of finite index. Then, for each (closed) neighborhood U of id_Y, there is $\beta \in (\Gamma')^\omega$ and $s \in S$ such that $\beta^{-1}\Gamma'\beta \subseteq \Gamma^{(R)}$ and $l\left(\beta^{-1} s \beta\right) \in U \setminus \{\text{id}_Y\}$.

Proof. For any $\gamma \in \Gamma^\omega$ and any $0 < r \in \mathbb{R}^*$, let

$$\delta(\gamma, r) = \max\{d(\gamma a, a) : \ell(a) \leq r\}.$$

We claim that

$$\delta\left(g^{-1}\gamma g, r\right) \leq \delta(\gamma, r) + 2\ell(g).$$

Suppose $\ell(a) \leq r$. Then, by the left-invariance of d, we have

$$\begin{aligned} d\left(g^{-1}\gamma g a, a\right) &= d(\gamma(ga), ga) \\ &\leq \delta(\gamma, \ell(g) + r) \\ &\leq \delta(\gamma, r) + 2\ell(g), \end{aligned}$$

where the latter inequality emerges from the fact that if we write an element of $B(e, \ell(g) + r)$ as bc for some

$b \leq r$ and $c \leq \ell(g)$, then

$$\begin{aligned}\delta(\gamma bc, bc) &\leq d(\gamma bc, \gamma b) + d(\gamma b, b) + d(b, bc)\\ &\leq d(bc, b) + d(\gamma b, b) + d(bc, b)\\ &\leq \delta(\gamma, r) + 2\ell(g).\end{aligned}$$

Now, let $U = U(k, \varepsilon)$ be a fixed neighborhood id for some $k \in \mathbb{N}$ and $\varepsilon > 0$. Translating the above lemma into nonstandard language yields that we have some $s \in S$ and $g \in (\Gamma')^{\omega}$ such that $\ell\left(g^{-1} s g\right) > \varepsilon R$ for some infinite $R \in \mathbb{R}^*$.

Write $g = s_1 \cdots s_t$ for some nonstandard natural $t \in \mathbb{N}^*$. For any $0 \leq i \leq t$, write

$$\begin{aligned}g_i &= s_1 \cdots s_i\\ M_i &= \max\left\{\delta\left(g_i^{-1} s g_i, kR\right) : s \in S\right\}.\end{aligned}$$

Let $C = \max\{\ell(s) : s \in S\}$, meaning that $C \in \mathbb{N}$. Notice that we have $M_0 < \varepsilon R$ since Γ' acts trivially on Y, and $M_t > \varepsilon R$ by the assumption. Furthermore, we have $|M_{i+1} - M_i| < 2C$ for each $0 \leq i \leq t-1$. Therefore, there is some $0 \leq i \leq t$ such that $|M_i - \varepsilon R| < 2C$. Define $\beta = g_i$, so that $\beta \in (\Gamma')^{\omega}$.

If $\gamma \in \Gamma'$, then $\beta^{-1}\gamma\beta$ is a finite product of elements of the form $\beta^{-1} s \beta$ with $s \in S$, each of which is an element of $\Gamma^{(R)}$ by how we defined β. Thus, $\beta^{-1}\Gamma'\beta \subseteq \Gamma^{(R)}$. In particular, we obtain the existence of some $s \in S$ such that $\delta\left(\beta^{-1} s \beta, kR\right)$ differs from εR by at most $2c$. If $\sigma = l\left(\beta^{-1} s \beta\right)$, then $\sigma \neq \mathrm{id}$, and for any $\ell(a) \leq kR$, we have that

$$\begin{aligned}d(\sigma(aN), aN) &= \mathrm{st}(d(\sigma(a), a)/R)\\ &\leq \varepsilon,\end{aligned}$$

so that $\sigma \in U$. Thus, we obtain our desired result. □

Thus, we obtain the following cases.

If $l(\Gamma)$ is infinite, then we have a finite index subgroup $\Delta \subseteq \Gamma$ such that we have a nontrivial homomorphism $\Delta \to \mathbb{Z}$, so by the process of proving Lemma 1, it follows that Γ virtually nilpotent.

If $l(\Gamma)$ is finite, but Γ has an abelian subgroup of finite index, then we are done since abelian groups are nilpotent.

If $l(\Gamma)$ is finite, then $\Gamma' = \ker(\Gamma)$ has finite index in Γ. By the above proposition, Γ' has homomorphic images in $\mathrm{Isom}(Y)$ containing elements not equal to id that are arbitrarily close to id. Since $\mathrm{Isom}(Y)$ is a Lie group, it has the No Small Subgroups property, meaning that for each $n > 0$, there is a neighborhood of id containing no elements of order less than or equal to n. Thus, Γ' has arbitrarily large homomorphic images in $\mathrm{Isom}(Y)$, so an application of Lemma 1, along with the fact that a finite index subgroup of a finite index subgroup of Γ is finite index in Γ, once again yields our desired result.

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